

1 0. Prof's Blueprint (High Priority)

- **Errors:** abs/rel error, MAE, RMSE on discrete points.
- **Polynomial forms:** standard vs nested (Horner) vs centered; FLOP counts.
- **Horner:** evaluate $p(c)$; synthetic division by $ax - b$; derivative via repeated Horner.
- **Taylor:** construct poly at given center; truncation error integral & Lagrange form; bound on interval (Sandwich Thm); find n for tolerance; Maclaurin/Taylor series convergence.
- **Interpolation:** setup linear system + matrix form $Xa = y$; Vandermonde det (why nonzero); over/underdetermined & free parameters; Lagrange basis $\ell_i(x)$ & poly $p(x)$; extrapolation vs interpolation; divided differences (1st/2nd order only, no full Newton); Chebyshev nodes, Runge phenomenon; Bézier curves (basic).
- MCQ + written section, 30 pts total. Bring 1 A4 cheat sheet (this one!).

2 1. Purpose of NA & Solution Types (Th.1–2)

Why numerical methods? Exact answers often *unavailable or impractical*: (a) no closed-form expression exists (e.g. $\int e^{-x^2} dx$, $\sin x = x$); (b) formula exists but is too slow/unstable to evaluate; (c) data only known at discrete measured points. **Definitions:** **exact solution** = satisfies equation exactly (symbolic); **closed-form/analytical** = explicit expression in known functions/constants; **approximate (numerical)** = value from an algorithm with controlled error, generally not equal to exact. **Locality:** every approximation is built to be good *near* a reference point/interval — Taylor about x_0 , interpolation on nodes $[x_0, x_n]$. Quality degrades away from that neighborhood $\Rightarrow$ **extrapolation is unreliable**.

3 2. Error Formulas

$$E_{abs} = |f(x) - \tilde{f}(x)|, \quad E_{rel} = \frac{|f(x) - \tilde{f}(x)|}{|f(x)|}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |f(x_i) - \tilde{f}(x_i)|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (f(x_i) - \tilde{f}(x_i))^2}$$

Trap: RMSE penalizes large errors more (squares); MAE treats all equally. Round to 5 dp unless told otherwise.

4 3. Polynomial Forms

Standard: $p_n(x) = a_n x^n + \dots + a_1 x + a_0$ — needs $\sim 2n - 1$ mults, n adds (naive) or $n(n+1)/2$ mults w/o precomputed powers.

Nested (Horner): $p_n(x) = (\dots((a_n x + a_{n-1})x + a_{n-2})x + \dots)x + a_0$ — n mults, n adds. **Always fewer ops.**

Centered at x_0 : $p_n(x) = \sum b_k (x - x_0)^k$. Get b_k via repeated synthetic division by $(x - x_0)$, or Taylor: $b_k = p^{(k)}(x_0)/k!$.

Horner's Method (synthetic division)

Divide $p(x)$ by $(x - c)$: bring down a_n , each next coeff = $a_k + c \cdot$ (prev result). Last value = remainder = $p(c)$.

	a_n	a_{n-1}	$\dots$	a_0
c		cb_{n-1}	$\dots$	cb_0
	b_{n-1}	b_{n-2}	$\dots$	r

- $p(x) = (x - c)q(x) + r$, $r = p(c)$; bottom row (minus last) = coeffs of $q(x)$.
- Divide by $(ax - b)$: divide by $(x - b/a)$ first (root $c = b/a$), get $q^*(x)$; true quotient $q(x) = q^*(x)/a$, same remainder r .
- **Derivative link:** run the tableau on $p(x)$ to get $q(x)$ & $r = p(c)$; run *another* tableau on $q(x)$ at same $c \Rightarrow$ new remainder = $p'(c)$. (Repeat on next quotient for $p''(c)/2!$, etc.)
- **Trap:** two ways to get $p'(1)$ — direct Horner on $p'(x)$, or double-tableau on $p(x)$ — must match.

5 4. Taylor & Maclaurin

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x)$$

Maclaurin: $x_0 = 0$.

Truncation Error $R_n(x)$

Integral form:

$$R_n(x) = \int_{x_0}^x \frac{(x-t)^n}{n!} f^{(n+1)}(t) dt$$

Lagrange form:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$$

for some ξ between x_0 and x . **Relationship:** Lagrange form = Mean Value Thm applied to the integral form (weighted avg of $f^{(n+1)}$ over interval).

Bounding $R_n(x)$ on interval $[a, b]$ (Sandwich Thm)

1. Find $M = \max_{\xi \in [a, b]} |f^{(n+1)}(\xi)|$ (often = bound on derivative amplitude, e.g. for $\sin, \cos$ terms $M \leq 3^{n+1}$ etc.).
2. Find max of $|x - x_0|^{n+1}$ on $[a, b]$ (largest distance from center).
3. $|R_n(x)| \leq \frac{M \cdot d^{n+1}}{(n+1)!}$ where $d = \max |x - x_0|$.

Finding n for tolerance $|R_n| < \varepsilon$

Set bound expression $< \varepsilon$, solve for smallest integer n (try increasing n until factorial growth beats d^{n+1} growth). Use $n!$ growth dominates exponential — trial/error with a table is acceptable.

"Largest x " type (Quiz 3 Q5 style)

Fix n ; treat bound $\frac{M d^{n+1}}{(n+1)!} < \varepsilon$ as inequality in x (since $d = x - x_0$ here); solve for x .

Series Convergence

Taylor series = $f(x) \iff \lim_{n \rightarrow \infty} R_n(x) = 0$. Writing the infinite sum from the pattern does *not* by itself prove convergence — must separately verify $R_n \rightarrow 0$ (true for $\sin, \cos, e^x$ on all $\mathbb{R}$ since M bounded, $d^{n+1}/(n+1)! \rightarrow 0$).

Maclaurin of $\sin(3x)$

$$\begin{aligned} \sin(3x) &= \sum_{k=0}^{\infty} \frac{(-1)^k (3x)^{2k+1}}{(2k+1)!} \\ &= 3x - \frac{27x^3}{6} + \frac{243x^5}{120} - \dots \end{aligned}$$

Trap: degree-8 Maclaurin poly of an odd function still stops at x^7 term (even-degree coeffs are 0) — write poly *through* degree 8, don't force an x^8 term.

6 5. Chain Rule / Derivative Practice

$[f(g(x))]' = f'(g(x))g'(x)$. Common combos: $2 \sin x + e^x$; $1 + \cos^2 x = 1 + (\cos x)^2 \Rightarrow$ derivative = $-2 \cos x \sin x = -\sin(2x)$. **Trap:** evaluate derivatives at the *center* x_0 before plugging into Taylor coeff $f^{(k)}(x_0)/k!$ — don't leave symbolic. Useful values: $\sin(\pi/2) = 1, \cos(\pi/2) = 0$; $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$; $\sin(-\pi/4) = -1/\sqrt{2}$.

7 6. Interpolation Basics

Interpolation: passes exactly through given data points (local fit). **Approximation:** fits overall trend, need not pass through points (global fit, e.g. least squares). **Extrapolation:** evaluate poly *outside* node range — much less reliable than interpolation (points 4-5 pattern in quizzes: always flag extrapolation as less trustworthy).

Degree-1 (Linear) Interpolation

Given $(x_0, y_0), (x_1, y_1)$: $p_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$.

Main Types of Interpolation (Th.8)

- **Linear (piecewise):** connect consecutive points by line segments. Simple, cheap, C^0 only (kinks at nodes). Good for monotone data; crude derivative info.
- **Global poly** (Lagrange/Newton/Vandermonde): one degree- n poly through $n+1$ pts; C^∞ ; **unique** for distinct nodes. Limit: high degree $\Rightarrow$ ill-conditioned, Runge.
- **Cubic splines:** piecewise cubic with matched f, f', f'' at nodes; C^2 smooth, local updates, avoids Runge — best for many nodes.
- **Hermite:** matches f and f' at nodes ($2n+2$ conditions $\Rightarrow$ degree $2n+1$).

Key tradeoff: more smoothness $\Rightarrow$ more conditions $\Rightarrow$ higher degree (less stable); local (spline) methods avoid Runge, global ones risk it.

Interpolation Error Bound

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

$$|f(x) - p_n(x)| \leq \frac{M}{(n+1)!} \max_x \left| \prod_{i=0}^n (x - x_i) \right|$$

where $\xi \in [x_{\min}, x_{\max}]$, $M = \max |f^{(n+1)}|$. **Uses:** bound error at a point; choose nodes to shrink product term — why Chebyshev (minimize $\max |\prod|$) beat uniform; error = 0 at every node. Analogous to Taylor remainder with $(x - x_0)^{n+1}$ replaced by $\prod (x - x_i)$.

Vandermonde System

For $n+1$ points, poly degree n : $a_0 + a_1 x_i + \dots + a_n x_i^n = y_i$ for each i . Matrix form $Xa = y$ where

$$X = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{pmatrix}$$

$$\det(X) = \prod_{0 \leq i < j \leq n} (x_j - x_i)$$

Why nonzero: interpolation nodes x_i are *distinct* $\Rightarrow$ every factor $(x_j - x_i) \neq 0 \Rightarrow \det \neq 0 \Rightarrow$ unique solution guaranteed.

Free Parameters (over/underdetermined)

points = $m+1$, poly degree = n . If $n > m$: **underdetermined** — infinitely many interpolating polys; free parameters = $n - m$ (extra coefficients not pinned down by the $m+1$ equations). If $n < m$: **overdetermined** — generally no exact solution (unless data is consistent) $\rightarrow$ use approximation (least squares) instead. **Trap:** "degree at most n " with only $m+1 < n+1$ points $\Rightarrow$ family has $n-m$ free parameters — identify them explicitly as the highest-order coefficients.

Lagrange Interpolation

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \quad p(x) = \sum_{i=0}^n y_i \ell_i(x)$$

Properties: $\ell_i(x_j) = \delta_{ij}$ (1 if $i = j$, else 0); basis *depends only on nodes x_i* , not on y_i or f — same ℓ_i reused for any function on same nodes. **Why a basis (Th.12):** $\ell_0, \dots, \ell_n$ are $n+1$ polys of degree $\leq n$ and are *linearly independent* (a combo $\sum c_i \ell_i \equiv 0$ evaluated at node x_j gives $c_j = 0$). Since $\dim P_n = n+1$, they span all polys of degree $\leq n \Rightarrow$ they form a basis; any data (x_i, y_i) yields unique coefficients $c_i = y_i$. **Steps:** (1) list nodes; (2) build each $\ell_i(x)$ as product of $(x - x_j)/(x_i - x_j)$ over $j \neq i$; (3) sum $y_i \ell_i(x)$; (4) simplify/evaluate at target point; (5) compute abs/rel error vs known true value if given; (6) extrapolate = plug x outside node range into same $p(x)$.

8 7. Divided Differences & Newton (Th.13)

0th order: $f[x_i] = f(x_i)$ (the node value itself).

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

General (order k): $f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$, denom = last — first node.

Newton Interpolation Formula (Th.13)

$$p_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j)$$

Advantages vs Lagrange: adding a node adds only *one* term (no basis recompute); nested (Horner-like) eval cheaper; incremental DD table. Lagrange pros: no table, basis reusable for *any* f on same nodes. (Per prof: quizzes need only 1st/2nd order DD; know formula & why Newton wins for incremental data.)

9 8. Grids, Runge, Chebyshev, Bézier

- **Uniform grid:** equally spaced nodes, $x_i = a + ih$, $h = (b - a)/n$. **Non-uniform:** unequal spacing (e.g. Chebyshev).
- **Runge's phenomenon:** high-degree interpolation on *uniform* nodes over $[-1, 1]$ -type intervals can oscillate wildly near endpoints as n increases. Fix: Chebyshev nodes or splines.
- **Chebyshev nodes** (roots of T_{n+1}), on $[-1, 1]$:

$$x_k = \cos\left(\frac{2k+1}{2(n+1)}\pi\right), \quad k = 0, \dots, n$$

- On $[a, b]$: $x_k = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{2k+1}{2(n+1)}\pi\right)$. Cluster near endpoints; minimize max of $|\prod_{i=0}^n (x - x_i)|$ (interpolation error term) $\Rightarrow$ minimize Runge oscillation.
- **Bézier curves:** defined by control points; curve passes through *endpoint* control points only (not interior ones); used in computer graphics for smooth curve design; based on Bernstein polynomials as blending functions.

10 9. Worked-Pattern Checklists

Error Question Checklist

☐ compute $f(x_i)$ exactly ☐ compute $\tilde{f}(x_i)$ ☐ abs error = $|diff|$ ☐ rel error = $abs/|f|$ ☐ if multiple points: MAE (avg of abs), RMSE (sqrt avg of sq).

Horner Question Checklist

☐ write coeffs $a_n, \dots, a_0$ (**include zeros for missing powers!**) ☐ build synthetic division table (bring down, multiply by c , add) ☐ last entry = $p(c)$ ☐ row above = coeffs of $q(x)$ ☐ repeat table on $q(x)$ at same c for $p'(c)$ if asked ☐ for divisor $ax - b$: use $c = b/a$, remember to divide quotient coeffs by a .

Taylor Construction Checklist

☐ pick center x_0 ☐ compute derivatives $f, f', f'', \dots$ at x_0 up to needed order ☐ divide each by $k!$ ☐ assemble $\sum \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$ ☐ do *not* simplify away the $(x - x_0)^k$ factors unless asked to expand.

Truncation Error Bound Checklist

☐ identify $f^{(n+1)}(x)$ pattern (for sin, cos combos it's bounded by amplitude) ☐ find $M = \max|f^{(n+1)}|$ on given interval ☐ find $\max|x - x_0| = d$ on interval ☐ plug into $Md^{n+1}/(n+1)!$ ☐ for "find n ": increase n until bound $< \varepsilon$, show smallest such n ☐ for "find x ": solve bound $< \varepsilon$ for x algebraically or numerically.

Interpolation System Checklist

☐ count points $m + 1$ vs desired degree n ☐ write $m+1$ equations $a_0 + a_1x_i + \dots + a_nx_i^n = y_i$ ☐ arrange as $Xa = y$ (rows=points, cols=powers 0.. n) ☐ if square: compute Vandermonde det via product formula ☐ if not square: state under/overdetermined & # free params.

Lagrange Checklist

☐ list all nodes ☐ for each needed ℓ_i : multiply $(x - x_j)/(x_i - x_j)$ over all $j \neq i$ (don't forget *every* other node, incl. non-adjacent ones) ☐ sum $y_i\ell_i(x)$ for full $p(x)$ ☐ evaluate at target, compare to true value for errors ☐ extrapolate = same formula outside range, note reduced reliability.

11 10. Common Traps Summary

- Forgetting **zero coefficients** for missing powers in Horner tables.
- Confusing **quotient** $q(x)$ degree (always one less than $p(x)$) after synthetic division.
- Sign errors in $(x - x_0)$ vs $(x_0 - x)$ in centered form / Taylor.
- Odd/even function Maclaurin series — half the coefficients vanish; don't invent nonzero terms.
- Mixing up **integral** vs **Lagrange** form of R_n — integral needs no unknown ξ , Lagrange does.
- Free parameter count: it's (poly degree +1) - (# data points), not degree itself.
- Lagrange basis $\ell_i(x_j) = 0$ for $j \neq i$, $= 1$ for $j = i$ — sanity check your basis polys.
- Vandermonde det = 0 *only* if two nodes coincide — never zero for distinct interpolation nodes.
- Extrapolation error can be large — always flag lower confidence vs interpolation.
- Degree of sum of two polys = $\max(\deg)$; degree of product = $\text{sum}(\deg)$; watch for **cancellation** making sum degree drop (find α s.t. leading terms cancel).
- FLOP counts: nested form always $\leq$ standard form; state exact counts (n mult + n add for degree n nested).

- Round only final numeric answers to 5 decimals; keep exact fractions/symbols mid-derivation.

12 11. Quick Reference Values

$e \approx 2.71828$, $\pi \approx 3.14159$, $\sqrt{2} \approx 1.41421$, $1/\sqrt{2} \approx 0.70711$. $n!$: $5! = 120$, $6! = 720$, $7! = 5040$, $8! = 40320$, $9! = 362880$, $10! = 3628800$.

13 12. Worked Mini-Examples

Horner: $p(x) = 2x^3 - 3x^2 + 0x + 5$ at $c = 2$
Coeffs: 2, -3, 0, 5.

	2	-3	0	5
2	2	4	2	4
	2	1	2	9

So $p(2) = 9$; quotient $q(x) = 2x^2 + x + 2$; $p(x) = (x - 2)(2x^2 + x + 2) + 9$. **For** $p'(2)$: run another tableau on q 's coeffs 2, 1, 2 at $c = 2$:

	2	1	2
2	2	4	10
	2	5	12

So $p'(2) = 12$.

Lagrange: nodes (0, 1), (1, 3), (2, 7), find $p(1.5)$

$$\ell_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}$$

$$\ell_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = \frac{x(x-2)}{-1}$$

$$\ell_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}$$

$p(x) = 1 \cdot \ell_0(x) + 3 \cdot \ell_1(x) + 7 \cdot \ell_2(x)$. At $x = 1.5$: $\ell_0 = \frac{(0.5)(-0.5)}{2} = -0.125$, $\ell_1 = \frac{(1.5)(-0.5)}{-1} = 0.75$, $\ell_2 = \frac{(1.5)(0.5)}{2} = 0.375$. $p(1.5) = 1(-0.125) + 3(0.75) + 7(0.375) = -0.125 + 2.25 + 2.625 = 4.75$.

Lagrange + errors: arctan (Quiz 5)

$f(x) = \arctan x$, nodes 0, 1, 3 ($y = 0, 0.785, 1.249$).

$$\ell_0(x) = \frac{(x-1)(x-3)}{(0-1)(0-3)} = \frac{(x-1)(x-3)}{3}$$

$$\ell_1(x) = \frac{x(x-3)}{(1-0)(1-3)} = \frac{x(x-3)}{-2}$$

$$\ell_2(x) = \frac{x(x-1)}{(3-0)(3-1)} = \frac{x(x-1)}{6}$$

$p(x) = 0\ell_0 + 0.785\ell_1 + 1.249\ell_2$. At $x = 2$: $\ell_1 = \frac{2(-1)}{-2} = 1$, $\ell_2 = \frac{2(1)}{6} = \frac{1}{3} \Rightarrow p(2) = 0.785 + \frac{1.249}{3} = 1.20133$.

Errors vs $\arctan(2) \approx 1.107$: $E_{abs} = |1.107 - 1.20133| = 0.09433$, $E_{rel} = \frac{0.09433}{1.107} = 0.0852$ (8.5%).

Extrapolate $x = -0.5$ (outside $[0, 3]$): $p(-0.5) = 0.785\ell_1(-0.5) + 1.249\ell_2(-0.5) = -0.53075$ (true $\arctan(-0.5) = -0.46365$ — note large error: extrapolation unreliable).

Divided Differences: nodes (0, 1), (1, 3), (2, 7)

$f[x_0] = 1$, $f[x_1] = 3$, $f[x_2] = 7$; $f[x_0, x_1] = \frac{3-1}{1-0} = 2$, $f[x_1, x_2] = \frac{7-3}{2-1} = 4$; $f[x_0, x_1, x_2] = \frac{4-2}{2-0} = 1$.

Newton form: $p_2(x) = 1 + 2x + 1 \cdot x(x - 1) = 1 + 2x + x^2 - x = x^2 + x + 1$. **Check:** same poly as Lagrange (uniqueness). At $x = 1.5$: $2.25 + 1.5 + 1 = 4.75$ ✓ matches.

Vandermonde det: nodes 0, 1, 2

$$\det(X) = \prod_{0 \leq i < j \leq 2} (x_j - x_i) = (1-0)(2-0)(2-1) = 1 \cdot 2 \cdot 1 = 2 \neq 0$$

Degree-1 Interp: $\{(2, 5), (5, -1)\}$ (Quiz 4 Q1)

$p_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0) = 5 + \frac{-1 - 5}{5 - 2}(x - 2) = 5 - 2(x - 2) = -2x + 9$. Check: $p_1(2) = 5$, $p_1(5) = -1$. ✓ (Also: slope $m = -2$ through both pts.)

Degree-4 System & det: 5 pts (Quiz 4 Q2)

Points $\{(0, 3), (1, 2), (2, 5), (4, 9), (7, 8)\}$, degree 4: $p_4(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$. **System** $Xa = y$, rows = points, cols = powers 0..4:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 4 & 16 & 64 & 256 \\ 1 & 7 & 49 & 343 & 2401 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 5 \\ 9 \\ 8 \end{pmatrix}$$

det: $\det X = \prod_{i < j} (x_j - x_i)$ over the 5 nodes (10 ordered pairs $i < j$):

$$\det X = (1)(2)(1)(4)(3)(2)(7)(6)(5)(3) = 30240 \neq 0$$

Solving: first eq gives $a_0 = 3$; back-substitute remaining 4 eqs (e.g. Gaussian elim) for $a_1..a_4$. Non-singular $\Rightarrow$ unique degree-4 poly.

Free Parameters: 2 points, degree ≤ 3 (Quiz 4 Q3)

Points (2, 5), (5, -1): $m + 1 = 2$, so $m = 1$; want degree $n = 3$ (4 coeffs a_0, a_1, a_2, a_3). Equations: 2. Free params = $n - m = 3 - 1 = 2 = a_2, a_3$ free. **Construct the family:** constraints $a_0 + 2a_1 + 4a_2 + 8a_3 = 5$ and $a_0 + 5a_1 + 25a_2 + 125a_3 = -1$. Subtract: $3a_1 + 21a_2 + 117a_3 = -6 \Rightarrow a_1 = -2 - 7a_2 - 39a_3$; then $a_0 = 9 + 10a_2 + 70a_3$. **Family:** $p(x) = (9 + 10a_2 + 70a_3) + (-2 - 7a_2 - 39a_3)x + a_2x^2 + a_3x^3$, any a_2, a_3 . Check: $p(2) = 5$, $p(5) = -1 \forall a_2, a_3$. □

Error calc: $\sin x \approx x$ (Quiz 1)

Exact $f(x) = \sin x$, approx $\tilde{f}(x) = x$. At $x = 0.01$: $E_{abs} = |\sin(0.01) - 0.01| \approx |0.00999983 - 0.01| = 1.67 \times 10^{-7}$; $E_{rel} = \frac{1.67 \times 10^{-7}}{\sin(0.01)} \approx 1.67 \times 10^{-5}$. **MAE** over $x = -0.01, 0.01, 0.03$: errors $1.67 \times 10^{-7}, 1.67 \times 10^{-7}, 4.50 \times 10^{-6}$; $\text{MAE} = \frac{1.67 + 1.67 + 45.0}{3} \times 10^{-7} \approx 1.61 \times 10^{-6}$. **RMSE** $= \sqrt{\frac{(1.67)^2 + (1.67)^2 + (45.0)^2}{3}} \times 10^{-7} \approx 2.60 \times 10^{-6}$ (larger than MAE — squares penalize the big error).

Truncation error bound example

$f(x) = \sin(3x)$, $n = 8$ (so $n+1 = 9$), interval $[-3, 4]$, $x_0 = 0$: $f^{(9)}(x) = \pm 3^9 \cos(3x)$ or $\pm 3^9 \sin(3x) \Rightarrow M = 3^9 = 19683$. $d = \max\{|-3|, |4|\} = 4$.

$$|R_8(x)| \leq \frac{19683 \cdot 4^9}{9!} = \frac{19683 \cdot 262144}{362880} \approx 14219$$

(Large bound is fine — the point is applying the formula correctly, not that it's tight.)

Diff./Integration Quick Review

Product rule: $(fg)' = f'g + fg'$. Quotient rule:

$$(f/g)' = \frac{f'g - fg'}{g^2}. \quad \text{Chain rule: } [f(g(x))]' = f'(g(x))g'(x).$$

Integration by parts: $\int u dv = uv - \int v du$. **Example:** $\int x \sin x dx$: let $u = x$, $dv = \sin x dx \Rightarrow du = dx, v = -\cos x$.

$$\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C$$

Centered Form: full worked example

$p(x) = x^3 - 2x^2 + 3x - 5$, center $x_0 = 2$. Repeatedly synthetic-divide by $(x - 2)$; each remainder is the next b_k (coefficient of $(x - x_0)^k$).

$$\begin{array}{l} 1, -2, 3, -5 \rightarrow 1, 0, 3 \mid b_0=1 \quad (\text{Div.1}) \\ 1, 0, 3 \rightarrow 1, 2 \mid b_1=7 \quad (\text{Div.2}) \\ 1, 2 \rightarrow 1 \mid b_2=4, b_3=1 \quad (\text{Div.3}) \end{array}$$

So $p(x) = 1 + 7(x - 2) + 4(x - 2)^2 + (x - 2)^3$. **Check:** expand back — matches original. □ Each division: bring down leading coeff, next = coeff + 2x prev; last value = remainder.

Horner: divide by $ax - b$ (worked)

Divide $4x^3 - 7x^2 + 5x + 8$ by $2x + 1$ (i.e. $a = 2, b = -1$, root $c = -\frac{1}{2}$).

	4	-7	5	8
$-\frac{1}{2}$		-2	4.5	-4.75
	4	-9	9.5	3.25

$q^*(x) = 4x^2 - 9x + 9.5$, remainder $r = 3.25$ (remainder unchanged by scaling). True quotient $q(x) = q^*(x)/2 = 2x^2 - 4.5x + 4.75$. **Check:** $(2x+1)(2x^2 - 4.5x + 4.75) + 3.25 = 4x^3 - 7x^2 + 5x + 8$. ✓

Bézier Curves: construction

Bernstein form, n control pts $P_0, \dots, P_n$, $t \in [0, 1]$:

$$B(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i P_i$$

Quadratic ($n=2$): $B(t) = (1-t)^2 P_0 + 2t(1-t) P_1 + t^2 P_2$. Cubic ($n=3$): add $3t^2(1-t)P_2'$... — use full Bernstein sum above. **Key facts:** curve passes through P_0 (at $t=0$) and P_n (at $t=1$) only; interior points *pull* the curve but aren't touched. **Worked example:** quadratic, $P_0 = (0, 0)$, $P_1 = (1, 2)$, $P_2 = (3, 0)$, find $B(0.5)$:

$$B(0.5) = 0.25(0, 0) + 2(0.5)(0.5)(1, 2) + 0.25(3, 0) = (0, 0) + (0.5, 1) + (0.75, 0) = (1.25, 1)$$